Making research resources in Life Sciences and Health interoperable*

Walter Baccinelli[1,*,†] and Vedran Kasalica[1,*,†]

[1] *Health-RI, Jaarbeursplein 6 3521 AL Utrecht, the Netherlands*

Abstract

Interoperability challenges continue to limit the adoption of FAIR data practices across the Dutch Life Sciences and Health (LSH) domain. To address this, a four-year project within the Thematic Digital Competence Center – Life Sciences and Health (TDCC-LSH), started in June 2025. The project combines community-driven alignment with technical coordination to support interoperable research data workflows. Key activities include semantic interoperability using an upper-level ontology, harmonisation of research tools through ontologies (e.g., EDAM), and collaboration with ELIXIR, Research Data Alliance working groups, and large-scale research infrastructures.

Keywords

Interoperability, FAIR

1. Introduction

Finding, accessing, combining, and reusing research data and tools across the Life Sciences and Health (LSH) domain remains a major challenge in the Netherlands. Although many high-quality, domain-specific initiatives exist, a lack of technical and organisational alignment hinders interoperability and limits the implementation of FAIR (Findable, Accessible, Interoperable, Reusable) data practices at scale.

2. Interoperability project

Within the Thematic Digital Competence Center – Life Sciences and Health (TDCC-LSH) hosted by Health-RI a 4-years project, funded by Open Science NL, has started in June 2025 with the goal of addressing these challenges by assembling a collective research data workflow framework to strengthen FAIR practices across the Dutch LSH ecosystem. The initiative focuses on aligning stakeholders and infrastructures through a combination of community building and technical coordination. Dedicated communities and workshops are organised to gather needs and foster collaboration around metadata, data, tools, infrastructures, policies, to make the resources available or under development in the Dutch research landscape interoperable.

---

1. *SWAT4HCLS 2026: Semantic Web Applications and Tools for Health Care and Life Sciences, March 23-26 2026, Amsterdam, the Netherlands*

[1*] Corresponding author.
[†] These authors contributed equally.
walter.baccinelli@health-ri.nl (W. Baccinelli); vedran.kasalica@health-ri.nl (V. Kasalica)
0000-0001-8888-4792 (W. Baccinelli); 0000-0002-0097-1056 (V. Kasalica)

On the technical side, we are leveraging Health-RI's Semantic Interoperability initiative to address the interoperability among different data standards in healthcare though the use of a unifying upper-level ontology that defines the precise meaning of key concepts already present in widely used standards.

We also aim to support the interoperability of the tools used during the research process. For this purpose, we are collaborating with existing initiatives such as ELIXIR and Research Data Alliance working groups like MaLDReTH and GORC. In this context, we are making use of tools annotation through EDAM ontologies to harmonize and facilitate the findability and evaluation of tools and the functionalities implemented, and we aim at co-creating guidelines to drive the developers building interoperable software solutions.

Through the collaboration with an initial set of Large Scale Research Infrastructures (LSRIs), the initial ground for aligning also metadata across multiple domains is being prepared. The mapping of different metadata standards toward ISA profiles and the analysis of the compliance of systems such as FAIRDOM-SEEK, FAIR Data Stations and FAIR Data Point with such profiles.

Acknowledgements

This works has been funded by Open Science NL under the program "Versterken lokale en thematische DCC's" with grantID https://doi.org/10.61686/FMRPB01518

Declaration on Generative AI

During the preparation of this work, the authors used Chat-GPT-4 and in order to: Grammar and spelling check. After using these tool the authors reviewed and edited the content as needed and take full responsibility for the publication's content.